

Carlos Moreno Rojas

Biomedical Engineering · Electrical Engineering · Research & Development

PROFILE

Biomedical and Electrical Engineer with experience in electronics, embedded systems, instrumentation, medical devices, FPGA, CAD, prototyping, signal and image processing, system integration and experimental validation.

PROFESSIONAL EXPERIENCE

Candel Medical Company

Professional internship in hardware, electronics and firmware for transcranial electrical stimulation.

Meldic Ltda.

Medical equipment experience and technical development projects.

SELECTED ENGINEERING PROJECTS

CYBATHLON 2024 — Ciervo UC

Electrical Area Leader. Development and integration of electronics for a robotic transfemoral prosthesis.

FPGA — Universal Testing Machine

Zybo Z10, Vivado and VHDL for force, displacement, deformation and speed signals.

Vscan Air Enclosure

CAD and prototyping for a portable ultrasound device accessory.

Organ Preservation Machine

Electromechanical redesign, fluid systems, control and miniaturization.

Lung Segmentation — U-Net

Medical image segmentation using deep learning.

TECHNICAL SKILLS

Electronics & Embedded Systems

Analog/digital electronics, PCB, ESP32, STM32, Arduino, ADC/DAC, firmware and communications.

FPGA & Digital Design

Zybo Z10, Vivado and VHDL.

Programming & Data

Python, C++, acquisition, signal processing and image processing.

Biomedical Engineering

Instrumentation, organ perfusion, preservation, neurostimulation, NIR fluorescence and validation.

CAD & Prototyping

Fusion 360, CAD, 3D printing, machining and electromechanical integration.

EDUCATION

MSc Biomedical Engineering — Pontificia Universidad Católica de Chile — 2025–2027

Electrical Engineering — Pontificia Universidad Católica de Chile — 2025–2026

Biomedical Engineering — Pontificia Universidad Católica de Chile — 2021–2024

LEADERSHIP

Biomedical Engineering Student Chapter UC

President, 2024–2025.

IEEE EMBS PUC Chile

Webmaster since 2024.

LANGUAGES

Spanish

Native.

English

B2 / ALTE 3.